

1 Solution — GIAM 1.1

Problem 8

1.1 Problem

Give an argument as to why the product of two rational numbers is again a rational.

1.2 The Proof

Let x and y be rational numbers. By the textbook's definition (Section 1.1),

$$\mathbb{Q} = \left\{ \frac{a}{b} : a, b \in \mathbb{Z}, b \neq 0 \right\},$$

so each is a quotient of integers with nonzero denominator:

$$x = \frac{a}{b}, \quad y = \frac{c}{d}, \quad a, b, c, d \in \mathbb{Z}, \quad b \neq 0, \quad d \neq 0.$$

Multiply:

$$x \cdot y = \frac{a}{b} \cdot \frac{c}{d} = \frac{ac}{bd}.$$

To conclude $xy \in \mathbb{Q}$ we must check that ac/bd is a quotient of integers with nonzero denominator. Three small observations:

- $ac \in \mathbb{Z}$ — integers are closed under multiplication.
- $bd \in \mathbb{Z}$ — same.
- $bd \neq 0$ — because $b \neq 0$ and $d \neq 0$, and the integers have *no zero divisors*: a product of nonzero integers is nonzero.

Therefore $xy = \frac{ac}{bd}$ is itself a quotient of integers with nonzero denominator, i.e. $xy \in \mathbb{Q}$.

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1.3 Worked Example

To make the abstract proof concrete: take $x = 4/6$ and $y = 15/8$. Then

$$x \cdot y = \frac{4}{6} \cdot \frac{15}{8} = \frac{4 \cdot 15}{6 \cdot 8} = \frac{60}{48}.$$

The numerator $60 \in \mathbb{Z}$ and the denominator $48 \in \mathbb{Z} \setminus \{0\}$, so $60/48 \in \mathbb{Q}$. (Reducing to lowest terms gives $5/4$, but this reduction isn't needed for the rationality conclusion.)

1.4 Remark — Where the Argument Bottoms Out

The proof rests on *three* properties of $\mathbb{Z}$:

1. Integers are closed under multiplication (for the numerator).
2. Integers are closed under multiplication (for the denominator).
3. Integers have no zero divisors (so $bd \neq 0$).

All three are part of saying $\mathbb{Z}$ is an **integral domain**. The closure of $\mathbb{Q}$ under multiplication is *inherited* from these properties — there is no extra magic at the rational level.

1.5 Remark — $\mathbb{Q}$ is a Field

The same kind of argument gives all the other arithmetic operations for free:

Operation	Formula in lowest-terms-free form	Result in $\mathbb{Q}$?
sum: $\frac{a}{b} + \frac{c}{d}$	$\frac{ad+bc}{bd}$	yes
difference: $\frac{a}{b} - \frac{c}{d}$	$\frac{ad-bc}{bd}$	yes
product: $\frac{a}{b} \cdot \frac{c}{d}$	$\frac{ac}{bd}$	yes (the focus of this problem)
quotient: $\frac{a}{b} \div \frac{c}{d}$	$\frac{ad}{bc}$, defined when $c \neq 0$	yes

Table 1.1

Each result is a quotient of integers (closed under $+$, $-$, $\times$) with nonzero denominator (no zero divisors), so the result lies in $\mathbb{Q}$. **Closure under all four arithmetic operat-**

ions is exactly what it means to call $\mathbb{Q}$ a field.

1.6 Remark — Lowest Terms Is Not Required

The proof works for *any* representation of x and y as quotients of integers — lowest-terms or not. The example $(4/6) \cdot (15/8) = 60/48$ shows this: neither input is in lowest terms, the output isn't either, but the conclusion " $\in \mathbb{Q}$ " holds regardless.

This robustness is by design: the definition

$$\mathbb{Q} = \left\{ \frac{a}{b} : a, b \in \mathbb{Z}, b \neq 0 \right\}$$

is *intentionally* permissive about representation. Two fractions a/b and c/d represent the same rational iff $ad = bc$; but each individual representation lives in $\mathbb{Q}$ on its own.

1.7 Remark — Comparison with $\mathbb{N}, \mathbb{Z}, \mathbb{R}$

How do the other “basic sets” fare with respect to the four arithmetic operations?

Set	+	−	×	÷ (nonzero divisor)	name
$\mathbb{N}$	yes	no	yes	no	commutative monoid (under $\times$)
$\mathbb{Z}$	yes	yes	yes	no	integral domain
$\mathbb{Q}$	yes	yes	yes	yes	field
$\mathbb{R}$	yes	yes	yes	yes	field (and complete)

Table 1.2

The leap from $\mathbb{Z}$ to $\mathbb{Q}$ adds closure under division — which is the key reason rationals exist as a separate concept beyond integers. The product-closure question (Problem 8) is the *easy* one; division-closure is the substantive one that distinguishes $\mathbb{Q}$ from $\mathbb{Z}$.

1.8 Remark — Generalization to “Field of Fractions”

The closure-under-multiplication argument *generalizes verbatim* to any integral domain R . The **field of fractions** of R is

$$\text{Frac}(R) = \left\{ \frac{a}{b} : a, b \in R, b \neq 0 \right\} / \sim,$$

where $\sim$ is the equivalence relation $a/b \sim c/d \iff ad = bc$. Closure under multiplication: $(a/b)(c/d) = ac/bd$ — the proof is *the same* as ours, using only:

- R closed under multiplication ($ac \in R, bd \in R$).
- R has no zero divisors ($bd \neq 0$ when $b, d \neq 0$).

Examples:

- $\text{Frac}(\mathbb{Z}) = \mathbb{Q}$ — our setting.
- $\text{Frac}(k[x]) = k(x)$ — the field of *rational functions* over a field k .
- $\text{Frac}(\mathbb{Z}[i]) = \mathbb{Q}(i)$ — Gaussian rationals.

In every case “product of two fractions over an integral domain is a fraction over the same integral domain” by the *same* one-line proof.

1.9 Remark — Why “No Zero Divisors” is Needed

Skip step 3 of the proof — i.e. weaken “integral domain” to “ring with zero divisors” — and the argument breaks. The classic counterexample is $R = \mathbb{Z}/6\mathbb{Z}$: here $2 \cdot 3 = 0$ in R , so we can construct “fractions” $1/2$ and $1/3$ whose product would have denominator $2 \cdot 3 = 0$ — undefined.

This is why “field of fractions” is *defined only* for integral domains. For general rings one has the more cumbersome “total ring of fractions,” which inverts only the non-zero-divisors. The integers are blessed: every nonzero integer is a non-zero-divisor, so we get the clean construction $\mathbb{Q}$.

1.10 Remark — A Single-Line “Symmetric” Statement

The whole proof of Problem 8 can be stated in a single symmetric sentence:

Multiplication of fractions multiplies numerators and multiplies denominators; both operations live inside $\mathbb{Z}$, and a product of nonzero integers is nonzero, so $\mathbb{Q}$ is closed under multiplication.

That sentence packs four pieces: (i) the definition of $\mathbb{Q}$ as a set of fractions, (ii) the formula for multiplying fractions, (iii) closure of $\mathbb{Z}$ under $\times$, (iv) absence of zero divisors in $\mathbb{Z}$. Each piece is required; remove any one and the conclusion can fail (as the $\mathbb{Z}/6\mathbb{Z}$ example shows). But put all four together and the conclusion is automatic.